

Homework 1 — Written Problems

Quantitative Reasoning II (INF 1220)

Due Monday, September 21, 11:59pm, on Populi

Complete these problems by hand. Scan and turn in your work to Populi by the due date.

1. A screening test

The fecal immunochemical test, or FIT, is an at-home test for colorectal cancer. In the United States it is offered every year to adults aged 45 to 75 who have no symptoms.

Pooling the studies of how well it performs: the test detects 79% of the cancers that are there, and correctly clears 94% of the people who do not have cancer. Among average-risk adults who get screened, about 2 in every 1,000 turn out to have colorectal cancer.

Start with 100,000 people screened.

1. How many have cancer, and how many do not?
2. Of those who have it, how many test positive?
3. Of those who do not have it, how many test positive anyway?
4. Someone tests positive. What is the probability they have cancer?
5. Now count up every person the test got right — the positives who do have cancer, plus the negatives who do not. Out of 100,000, how often was the test correct? Put that number next to your answer to 4 and explain how both can be true at once.
6. A positive FIT result is followed by a colonoscopy, not by a diagnosis. Use your answer to 4 to explain why, in two sentences, to someone who has not taken this course.

2. The same test, different patients

The NHS in England ran the same test on a different group: 9,822 patients whose doctors had already referred them because they had symptoms that can indicate colorectal cancer. In that group, 3.3% turned out to have it.

The test is unchanged: 79% sensitivity, 94% specificity.

1. Out of 100,000 patients like these, what is the probability that someone who tests positive has cancer?
2. Nothing about the test changed. Say in one sentence what did change, and why it moved the answer so far.

3. A manufacturer advertises a test as “94% accurate.” Using your work in problems 1 and 2, say what that claim leaves out.

3. Trip insurance

Trip insurance on a \$700 flight costs \$45. If you have to cancel, it refunds the full \$700. There is a 4% chance you have to cancel. (*The figures in this problem are made up for the sake of the example.*)

1. What is the expected value of buying the insurance?
2. On expected value alone, should you buy it?
3. Give one reason a person might reasonably buy it anyway. Your reason should not claim that the expected value calculation is wrong.

4. Mean and median

Nine flights left Austin for Chicago O’Hare on August 4, 2025. None of them was cancelled or diverted, so these nine are all of them. Here is how many minutes late each one arrived, with negative numbers meaning it arrived early:

-12 -9 -6 -2 1 7 16 25 258

1. Compute the mean and the median.
2. One passenger says the day’s flights to Chicago averaged over half an hour late. Another says the typical flight was on time. Both are using this data correctly. Explain how.
3. Remove the 258 and recompute both. Which one moved more, and why?
4. You are writing one sentence for a news story about this route. Which number do you use, and what do you have to say alongside it to be honest?

5. Reading a real claim

The passage below is from Vanguard’s *How America Saves 2026*. Vanguard administers retirement plans for nearly five million workers and publishes this report every year. The two paragraphs are quoted in the order they appear. A *DC plan* is a defined contribution plan, the category that includes the 401(k).

Vanguard account balances are a measure of how much plan participants have accumulated for retirement with a given employer. In the U.S., DC plans are not a closed system. When participants change jobs or retire, their plan assets may remain with their former employer’s plan, be rolled over to another employer plan or an IRA, or be cashed out. As a result, current DC plan balances often do not reflect lifetime savings and are only a partial measure of retirement preparedness for most participants.

In 2025, the average account balance for Vanguard participants was \$167,970, while the median balance was \$44,115 ... The wide divergence between the average and median balance is due to a small number of very large accounts that significantly raised the average above the median.

One in four participants had an account balance of less than \$10,000, while 35% of participants had a balance of more than \$100,000 ... Because of this skewed distribution of assets, average balances are indicative of participants at about the 75th percentile — about 75% of participants had balances below the average, and 25% had balances above.

1. The average is roughly 3.8 times the median. Using Vanguard's own explanation, say why in your own words.
2. Vanguard says the average sits at about the 75th percentile. Explain what that means for a participant who has exactly the average balance.
3. Suppose a news story reports: *"The average American has \$167,970 saved for retirement."* What is wrong with that sentence?
4. Rewrite the sentence so that it is accurate. You may use either number.